# Design Justification File

SQRT (Sub-zero Quantitative Research Team)

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## Document History

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## Author Contributions

**JA** §1–4

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## Key Reference Documents

1. UCD-DRD-2026-0
2. SQT-PDR-MDD-0
3. SQT-PDR-SEP-0
4. SQT-PDR-PMP-0

# 1 Executive Summary

The SQRT mission is a high-altitude experimental demonstration designed to investigate the thermal and phase behaviour of water exposed to low-pressure and low-temperature environments, analogous to conditions associated with cryovolcanic plumes on icy moons. The experiment focuses on observing flash evaporation and subsequent condensation or freezing processes in a controlled near-space setting.

The payload is carried within a compact satellite constrained to a maximum volume of 2 liters and a total mass of 800 grams. The internal structure is divided into two isolated sections: an Electronics Bus, which houses all control electronics, power systems, communication hardware, and the water reservoir; and a vented Low-Pressure Chamber, where the experimental release and observation take place. This separation ensures the protection of sensitive electronics while allowing direct exposure of the water to the external environment.

At a predetermined altitude, a small volume of liquid water is released from an internal reservoir into the Low-Pressure Chamber using a servo-actuated pinch valve. The evolution of the water is monitored using cameras located in different positions to capture complementary views of the experiment. Preliminary mass and volume estimates indicate that the system meets all imposed constraints, demonstrating a simple, low-cost, and robust platform for studying phase-change phenomena in near-space conditions.

# 2 Mission Overview

**Mission Statement:** The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to answer the question of how water plumes on icy moons thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes will be done using a high-altitude demonstration. This will involve the sudden exposure of liquid water within the TupperSat to a colder, lower-pressure environment which will be generated by elevating the TupperSat system to an altitude of 20-25 km. At 25 km altitude, the average atmospheric pressure is about 25 mbar (2.5% of that at sea level) with a corresponding boiling point of water of approx 21$^{o}$C. To perform this exposure, an opening mechanism, consisting of a valve and a trigger sensor, will allow a small volume of water to "see" a colder, less pressured environment within the TupperSat.

## 2.1 Mission Goals

The SQRT mission has the following goals:

1. Expose a sample of water to the high-altitude environment at a set pressure/altitude.
2. Observe the temperature and phase of the water sample over time.
3. Communicate with the ground.

# 3 Design Overview

## 3.1 Design concept

The initial design concept consists of a cubic satellite limited to a 2 liters volume and 800 grams constraints. The dimensions and final mass are yet to be defined; they will be addressed in later stages of the process. Still, for the simulation's sketch the dimensions used are 20x12x8, resulting in 1.92 liters.

Ventilation holes are included to prevent structural damage due to pressure differentials during ascent.

This tupperware will house and protect all onboard electronics and payload components, as well as an internal structure to secure the allocation of the internal components.

Outside, an antenna is installed for communication purposes (as TupperSat recovery is not granted), and handlers are connected to the inner structure to attach to the balloon.

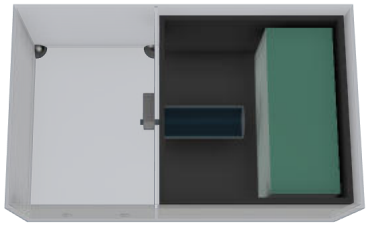


Figure 1: SQRT satellite initial design — above view

The internal display is divided into two sections for safety reasons, as the electronics can not be exposed and affected by the liquid. One of the sections, called Electronics Bus (right section) will contain all the electronics (green box) and the liquid container. It will define the operation; record and save the cameras' output; and send the data to ground, including the recorded data and the status of the valve. This Electronics Bus will be thermally insulated to maintain a proper operating temperature and the water in a liquid state.

The other section, called the Low-Pressure Chamber (left section), houses the actuator and cameras. In this section, the water will be ejected and tested. The actuator consists of the valve and servo-motor used to control the release of the liquid.

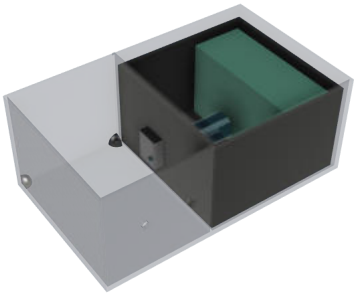
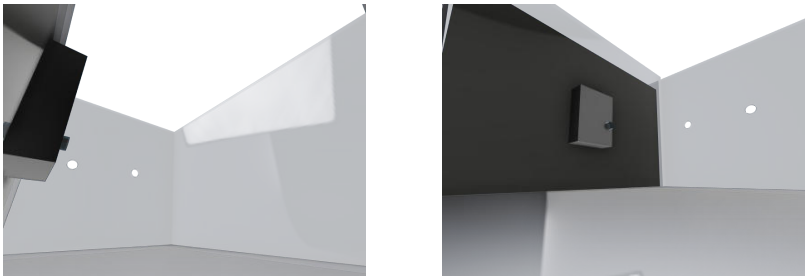


Figure 2: SQRT sat initial design corner view

For the initial design, two cameras are installed in the Low-Pressure Chamber to monitor the evolution of the water after valve opening. The camera model can operate in the Infrared or the optical spectrum, so the final location and configuration of each one will be addressed in the future. This configuration

allows for observing the process across different spectral ranges. The cameras are placed at different locations to provide complementary viewing angles and capture the full evolution of the experiment.



(a) (b)

Figure 3: Points of view of the two cameras. (a) View of the camera in front of the container. (b) View of the camera located at the corner of the container.

Given the environment the liquid will be in, the most suitable design is considered a cylinder with a conical mouthpiece to fit a plastic tube to orient the ejection and locate the servo valve.

## 3.2 Valve Options for Internal Water Release

The following valve options were considered for controlled water release from the internal reservoir into the experiment chamber:

- **Pinch valve**
  A flexible tube is mechanically compressed to block the flow and released to allow water discharge. This solution provides excellent sealing, high tolerance to vibration and freezing, and keeps the actuator isolated from the fluid.
- **Servo-actuated ball valve**
  A quarter-turn ball valve is rotated by a servo motor to open or close the flow path. This option offers a large flow area and clear open/close states but requires higher actuation torque and careful sealing.
- **Rupture membrane (burst disk)**
  A thin membrane seals the outlet and is intentionally punctured or ruptured to release the water. This approach ensures zero leakage during ascent but allows only a single, non-reversible release event.

The pinch valve was selected as the preferred option due to its simple design and reliable sealing performance, which are well suited to the vibration and rotation expected during ascent. Since the actuator does not come into direct contact with the water, the risk of leakage is reduced, and the system is less sensitive to partial freezing or evaporation.

Inside pinch valve designs, most commercially available options are based on solenoid actuation. However, these valves are generally too heavy for the mission constraints, with typical masses of around 200g for the lightest models, and their performance can be unreliable at low temperatures due to increased material stiffness and resistance.

For this reason, a servo-actuated pinch valve was selected and implemented using a 3D-printed mechanism, providing a lightweight, mechanically robust, and more adaptable solution for the expected flight conditions.

The selected servo-actuated pinch valve is fully reversible and allows repeated open–close cycles, enabling extensive ground testing before flight. Furthermore, once the water is ejected, the valve can be actuated again to seal the nozzle and maintain a controlled internal configuration.

# 4 Design Justification

## 4.1 Spatial allocation

The final container dimensions are still to be defined; the dimensions used in the model are preliminary (20x12x8, length x width x height). Still, the design will be inside the thresholds established for mass and volume: 800 grams and 2 liters.

To divide the two chambers with enough space for the water to spread in the Low-Pressure Chamber and the electronics to be safe and fit with the water container, a 3mm wall is located 81 mm from the external wall of the Low-Pressure Chamber.

Considering the optimal design, the liquid container is considered as a syringe of 15 ml, with the pusher at the bottom fixed. The dimensions are 20x50mm (diameter x height) without the pusher structure. It needs to be covered in thermal insulation material to maintain the water in a liquid state, as the outside temperature will be below 0ºC, and the tupperware has holes drilled for ventilation. The syringe can be located in a different spot to allow more space for the electronics if needed.

## 4.2 Water State Estimation

According to previous missions data, the temperature inside the Electronics Bus (insulated section) will be around 10ºC. At those altitudes (between 20 and 25 km), the temperature outside is around -50 °C and the pressure is between 50 and 25 mbar. According to the following diagram, the water will be in a liquid state in the container given the external pressure and internal temperature.

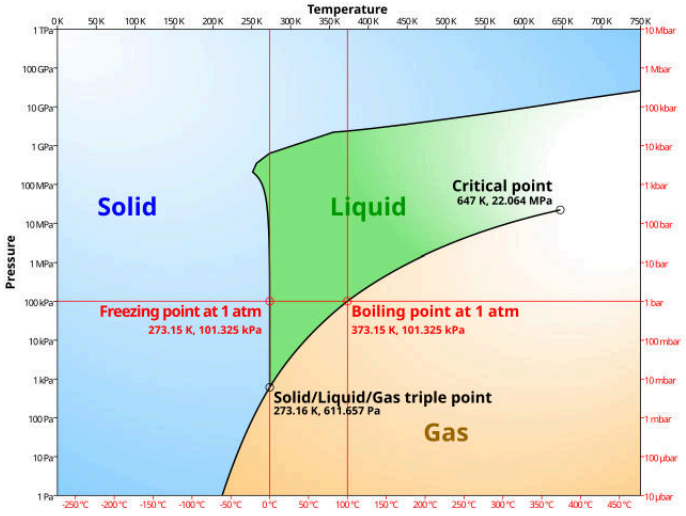



Figure 4: Simplified temperature/pressure phase change diagram for water

As the aim of the experiment is to observe a visible stream or spray during the release, the water needs to be at a higher temperature than the expected ambient conditions before the valve is opened. This is related to the behaviour of water under different temperature and pressure conditions, as shown in Figure 4.

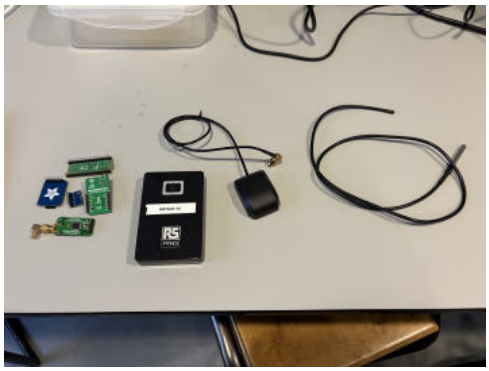
When the liquid is exposed to the low-pressure environment at high altitude, the sudden pressure drop should cause part of the water to evaporate. If the initial water temperature is relatively low (around 10ºC), the water is more likely to freeze quickly, with limited vapour being produced.

The water container will need to be held at a temperature over 25ºC. A heating pad will be used to ensure the temperature stays at least this high. The exposure of water is expected to produce a mixture

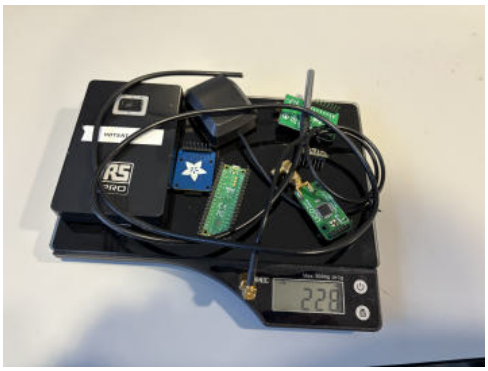
of vapour and liquid droplets before freezing occurs, increasing the chances of recording a clear and observable event.

## 4.3 Weight Estimation

For the materials given, the chips weigh 40g, the battery 130g, the GPS sensor 41g and the external temperature sensor 16g. That adds up to 228g.



(a)



(b)

Figure 5: (a) Main components given. (b) Weight of the components.

The Tupper container is estimated to weigh 200 g, leaving 370 g available for the structure, payload, and actuator. Using a 15 ml syringe results in approximately 15 g of water; including the syringe itself, the total payload mass is estimated to be 30 g.

The remaining components (i.e., cameras, valve, servo, and heating pad) are relatively lightweight. The structural elements (i.e., antenna, internal structure, and handling mechanism) are estimated to weigh 200 g. Any necessary materials or components not made in-house nor provided by UCD will be purchased using the budget outlined in SQT-PDR-PMP-0.

Table 1: Weight Estimation

| **Component** | **Estimated Weight (g)** |
|---|---|
| Chips | 40 |
| Battery | 130 |
| GPS sensor | 41 |
| Temperature sensors (x2) | 16 (x2) |
| Tupper container | 180 |
| Water (15 ml) + syringe | 30 |
| Insulation material | 40 |
| Heating pad | 6 |
| Camera (x2) | 3 (x2) |
| Actuator (servo valve + motor (9 g)) | 15 |
| Structure | 200 |
| **Total weight** | **720** |

With this total mass, a margin of 80 g remains to account for potential miscalculations or the inclusion of additional components.